



Dr Abdul Malik Tahir

Project Management

What is a Project?

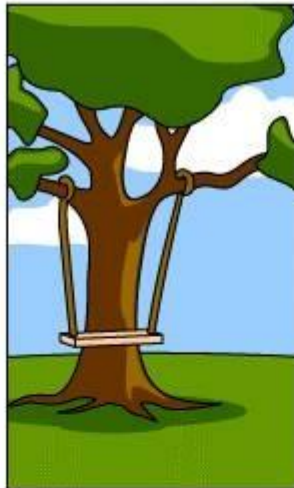
- A project is a temporary endeavor undertaken to produce a unique product or service



- **Temporary** – Definitive beginning and end
- **Unique** – New undertaking, unfamiliar ground



How the customer explained it



How the Project Leader understood it



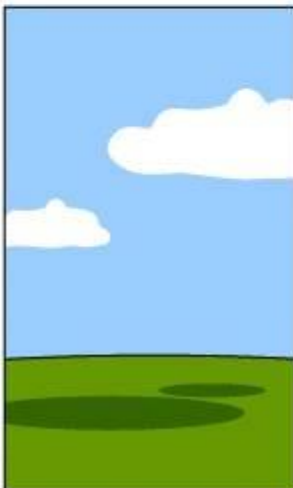
How the Analyst designed it



How the Programmer wrote it



How the Business Consultant described it



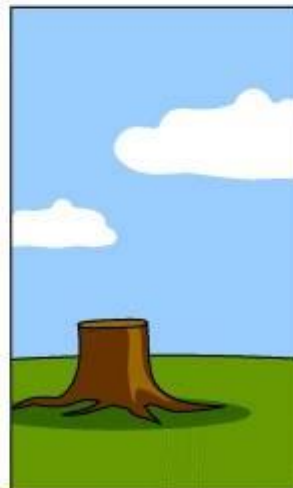
How the project was documented



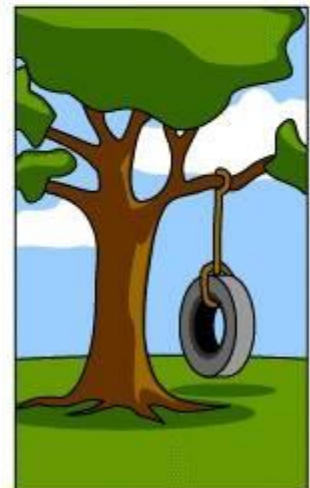
What operations installed



How the customer was billed



How it was supported



What the customer really needed

Project Success

**Customer Requirements
satisfied/exceeded**

**Completed within allocated
time frame**



**Completed within allocated
budget**

Accepted by the customer

Project Failure

Scope Creep

**Poor Requirements
Gathering**

**Unrealistic planning and
scheduling**

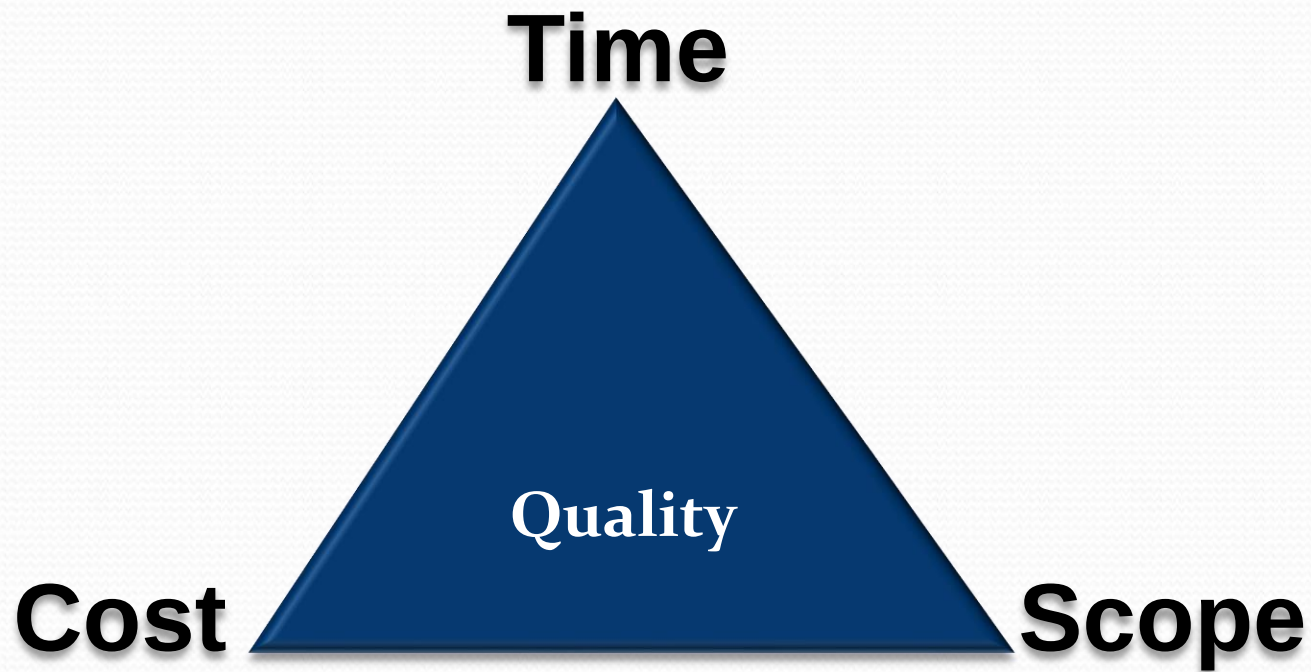
Lack of resources



What is Project Management

- Project Management is the application of skills, knowledge, tools and techniques to meet the needs and expectations of stakeholders for a project.
- The purpose of project management is **prediction** and **prevention**, NOT **recognition** and **reaction**

Triple Constraint



Triple Constraint

- Increased **Scope** = increased time + increased cost
- Tight **Time** = increased costs + reduced scope
- Tight **Budget** = increased time + reduced scope.

Key Areas of Project Management

- **Scope Management**
- **Issue Management**
- **Cost Management**
- **Quality Management**
- **Communications Management**
- **Risk Management**
- **Change Control Management**



Scope Management

- Primarily it is the definition and control of what *IS* and *IS NOT* included in the project.



Issue Management

- Issues are restraints to accomplishing the deliverables of the project.
- Typically identified throughout the project and logged and tracked through resolution.



Rope not thick

Issue... already impacting the cost, time or quality

Cost Management

- This process is required to ensure the project is completed within the approved budget and includes:



Resources

people
equipment
materials

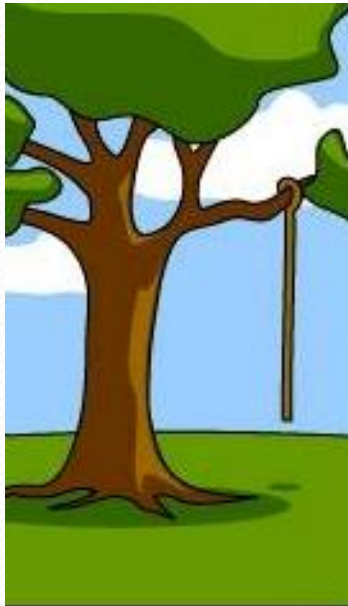
Quantities



Budget

Quality Management

- Quality Management is the process that insure the project will meet the needs



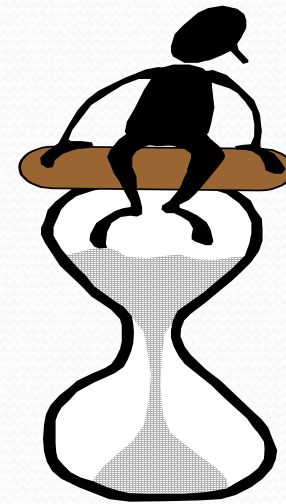
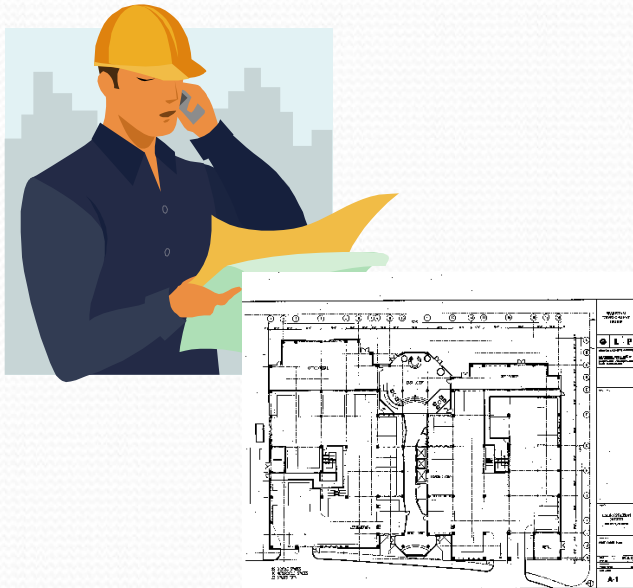
“conformance to requirements” - **Crosby**

“fitness for use” - **Juran**

“the totality of characteristics of an entity that bear on its ability to satisfy stated and implied need” - **ISO 8402:1994**

Communications Management

- This process is necessary to ensure timely and appropriate generation, collection, dissemination, and storage of project information



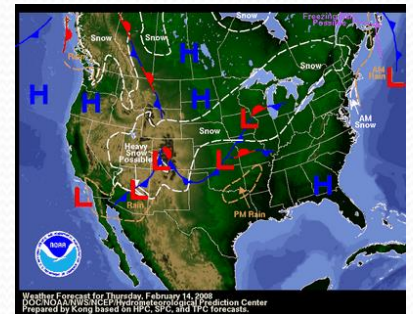
Risk Management

- Risk identification and mitigation strategy
- Risk update and tracking

Risk... POTENTIAL negative impact to project



Tree – location, accessibility,
ownership



Weather

Change Control Management

- Define how changes to the project scope will be executed

Scope Change



Technical Specification Changes



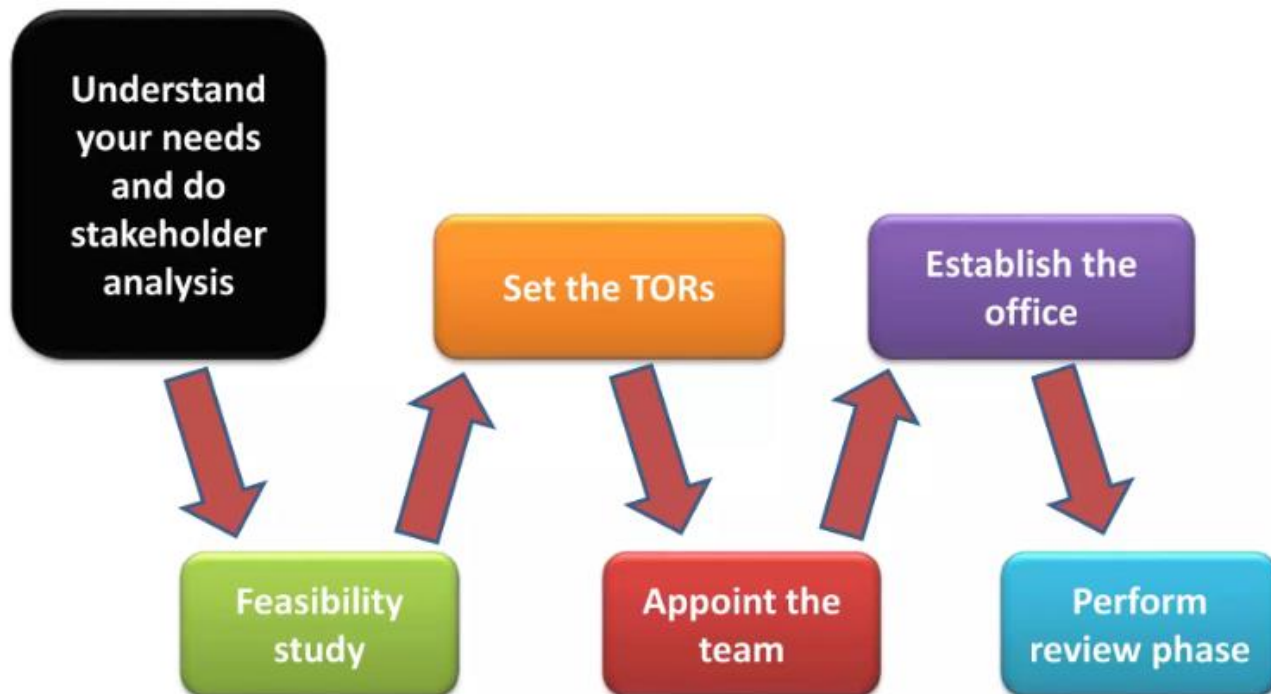
Schedule changes

All changes require collaboration and buy in via the project sponsor's signature prior to implementation of the changes

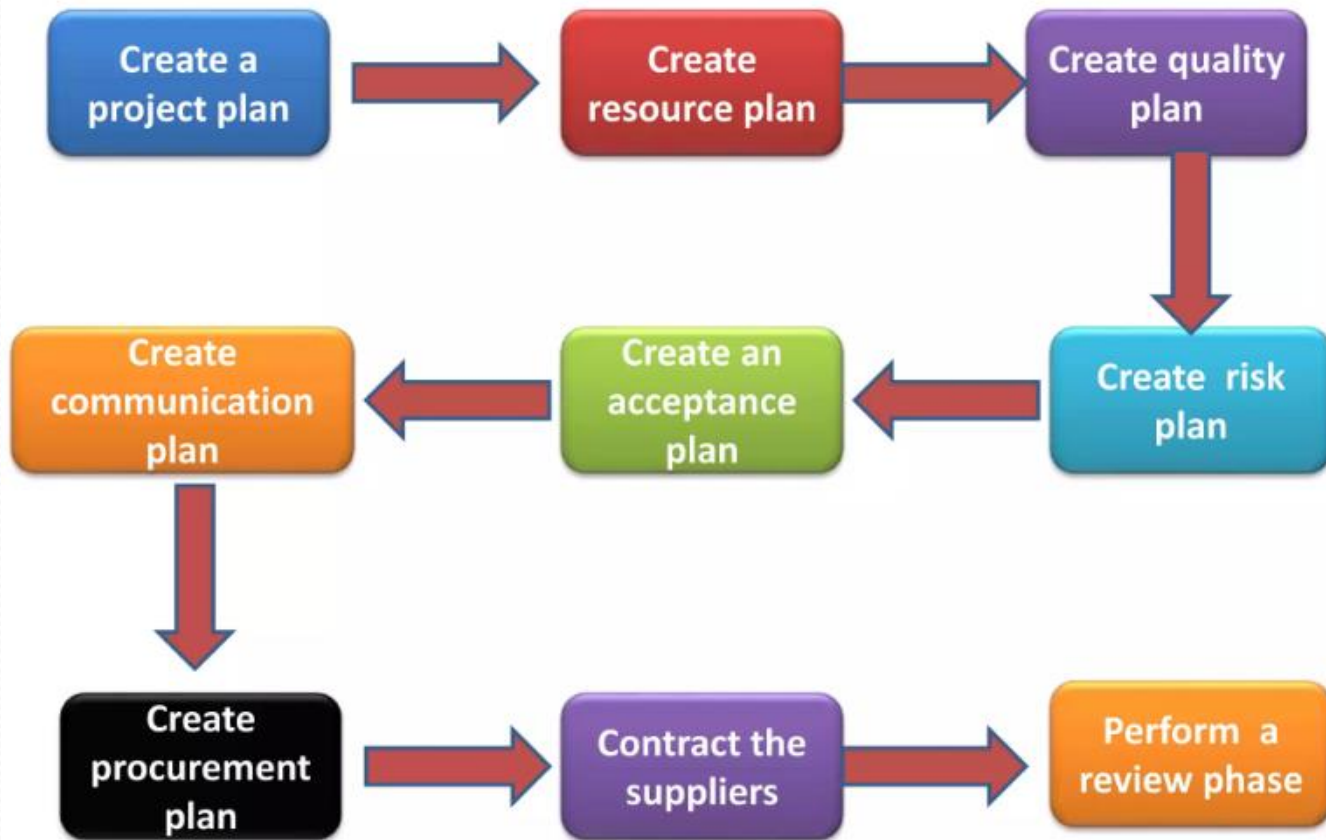
Five stages of Project Management



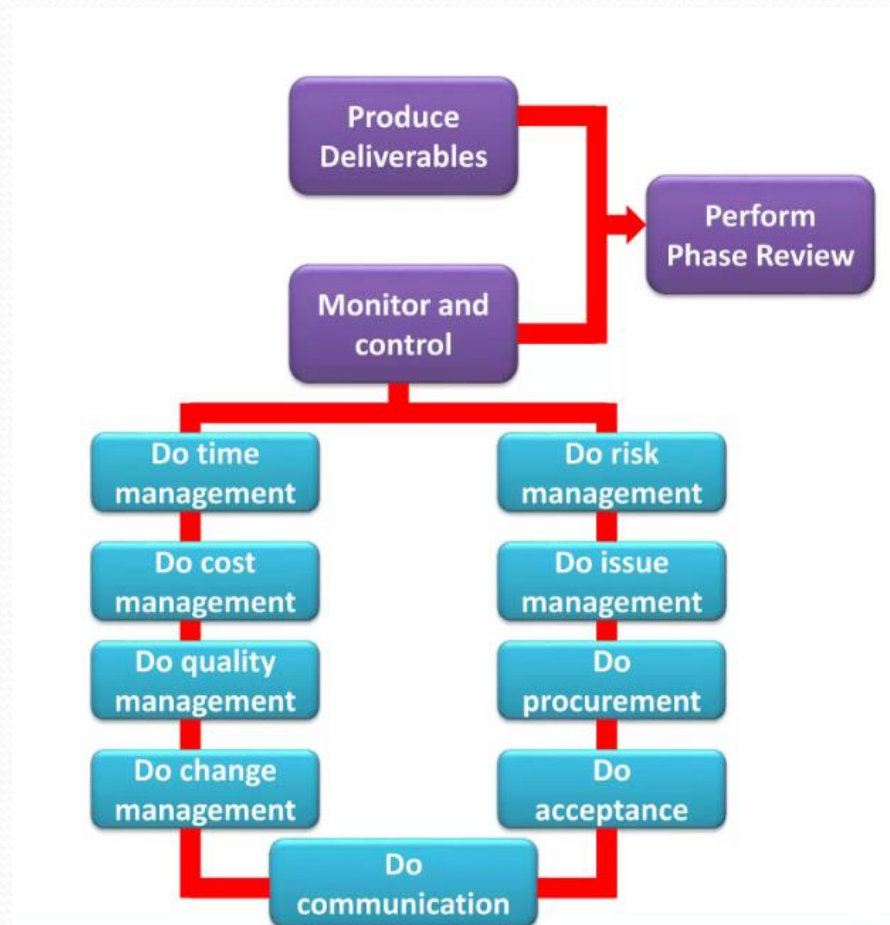
Initiation



Planing



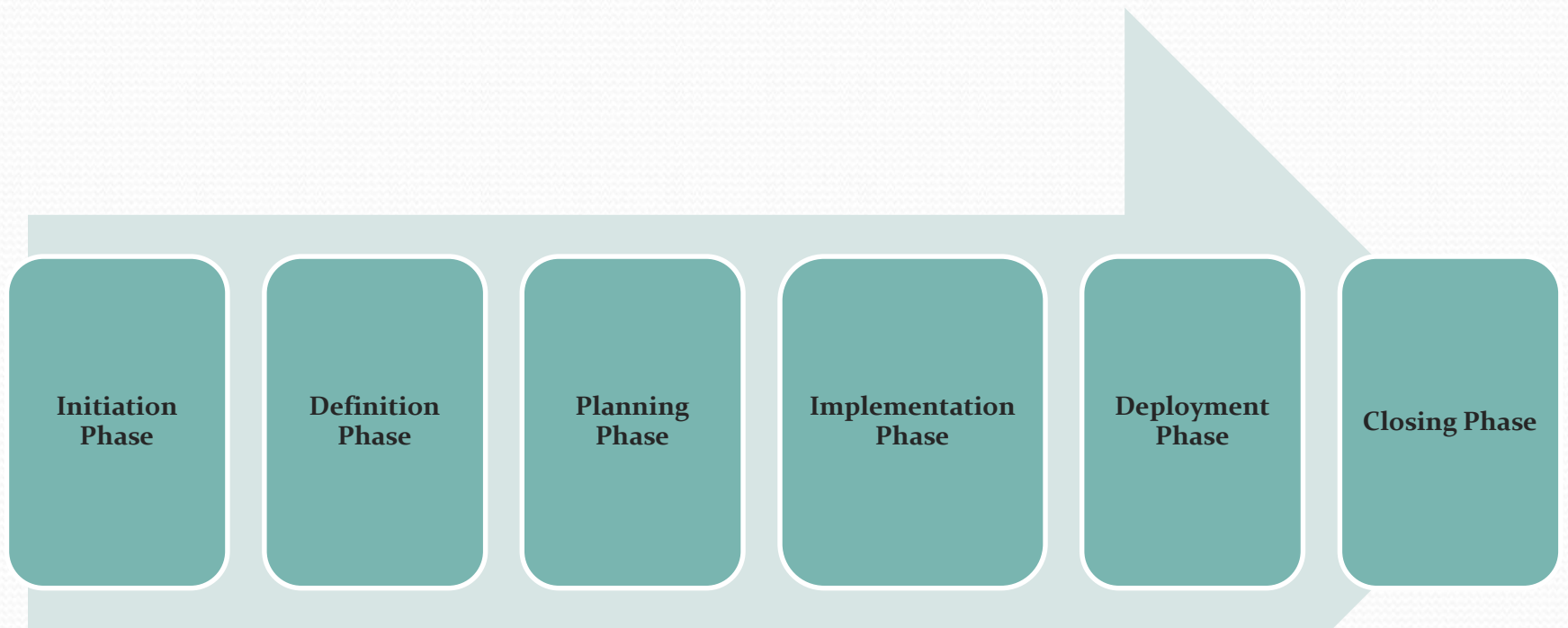
Execution



Top Reasons for Project Failure

- 84% - When serving on a project team, employees aren't relieved of some of their routine responsibilities
- 80% - Employees don't receive training in project management methodologies before serving on a project team.
- 69% - Project teams aren't given enough resources to accomplish their goals.
- 62% - Project teams throughout the organization fail to follow a standard methodology for defining, planning and implementing projects.
- 55% - The right people aren't selected to lead or serve on project teams.
- 46% - Project teams aren't given clear, attainable goals.

Project Life Cycle



Initiation Phase

- Define the need
- Return on Investment Analysis
- Make or Buy Decision
- Budget Development



Definition Phase

- Determine goals, scope and project constraints
- Identify members and their roles
- Define communication channels, methods, frequency and content
- Risk management planning



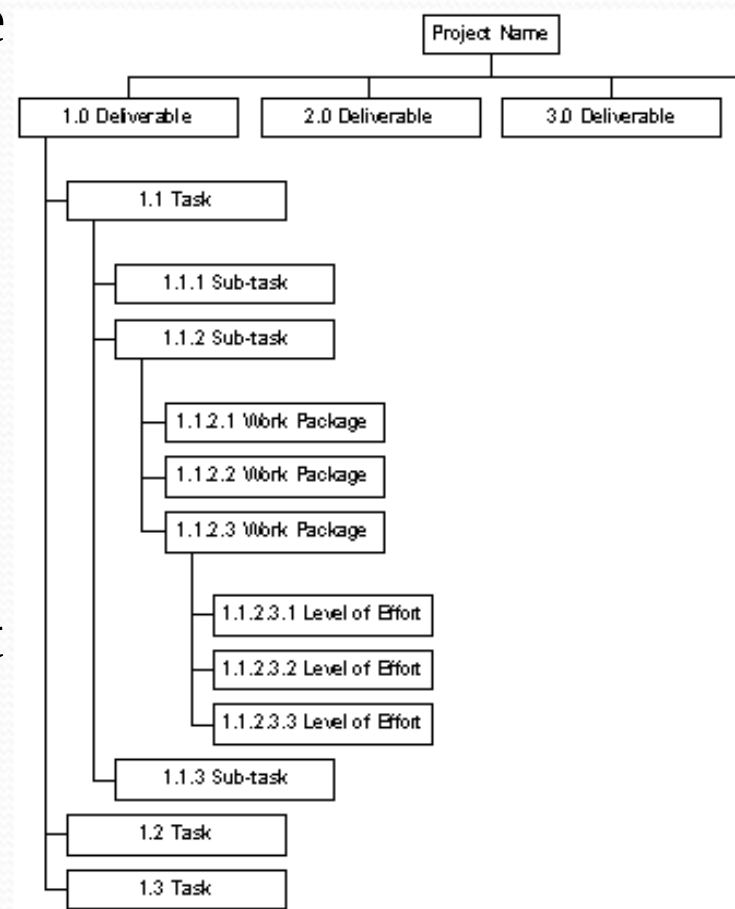
Planning Phase

- Resource Planning
- Work Breakdown Structure
- Project Schedule Development
- Quality Assurance Plan



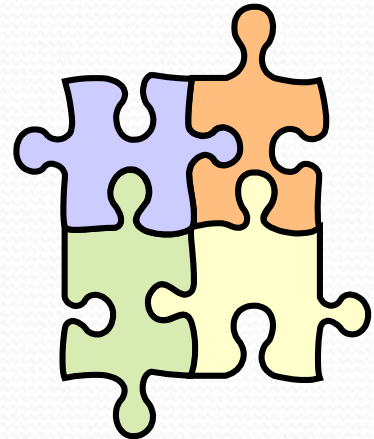
Work Breakdown Structure

- For defining and organizing the total scope of a project
- **First two levels** - define a set of planned outcomes that collectively and exclusively represent 100% of the project scope.
- **Subsequent levels** - represent 100% of the scope of their parent node



Implementation Phase

- Execute project plan and accomplish project goals
- Training Plan
- System Build
- Quality Assurance



Deployment Phase

- User Training
- Production Review
- Start Using



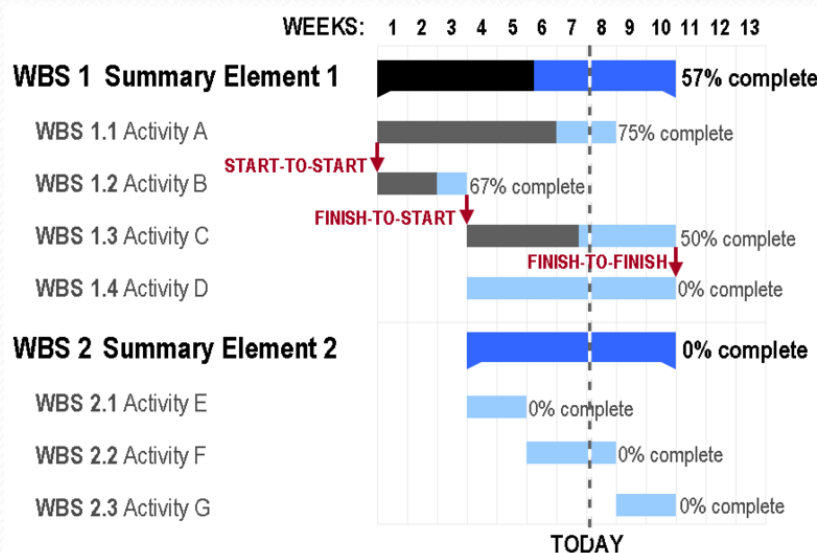
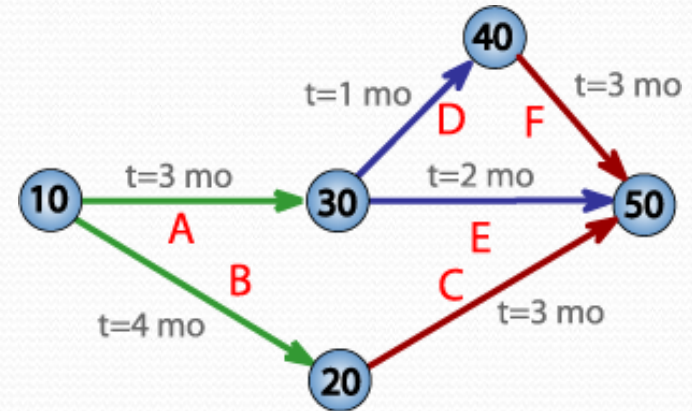
Closing Phase

- Contractual Closeout
- Post Production Transition
- Lessons Learned



Project Management Tools

- **PERT Chart**- designed to analyze and represent the tasks involved in completing a given project



- **Gantt Chart** - popular type of bar chart that illustrates a project schedule

Project Evaluation and Review Technique.

- A PERT chart, sometimes called a PERT diagram, is a project management tool used to schedule, organize and coordinate tasks within a project. It provides a graphical representation of a project's timeline that enables project managers to break down each individual task in the project for analysis.
- The PERT chart template uses nodes -- drawn as rectangles or circles -- to represent events and milestones throughout the project. The nodes are connected by vectors -- drawn as lines -- which represent the various tasks that need to be completed

- **Terminology**

Here is a list of terms associated with PERT charts:

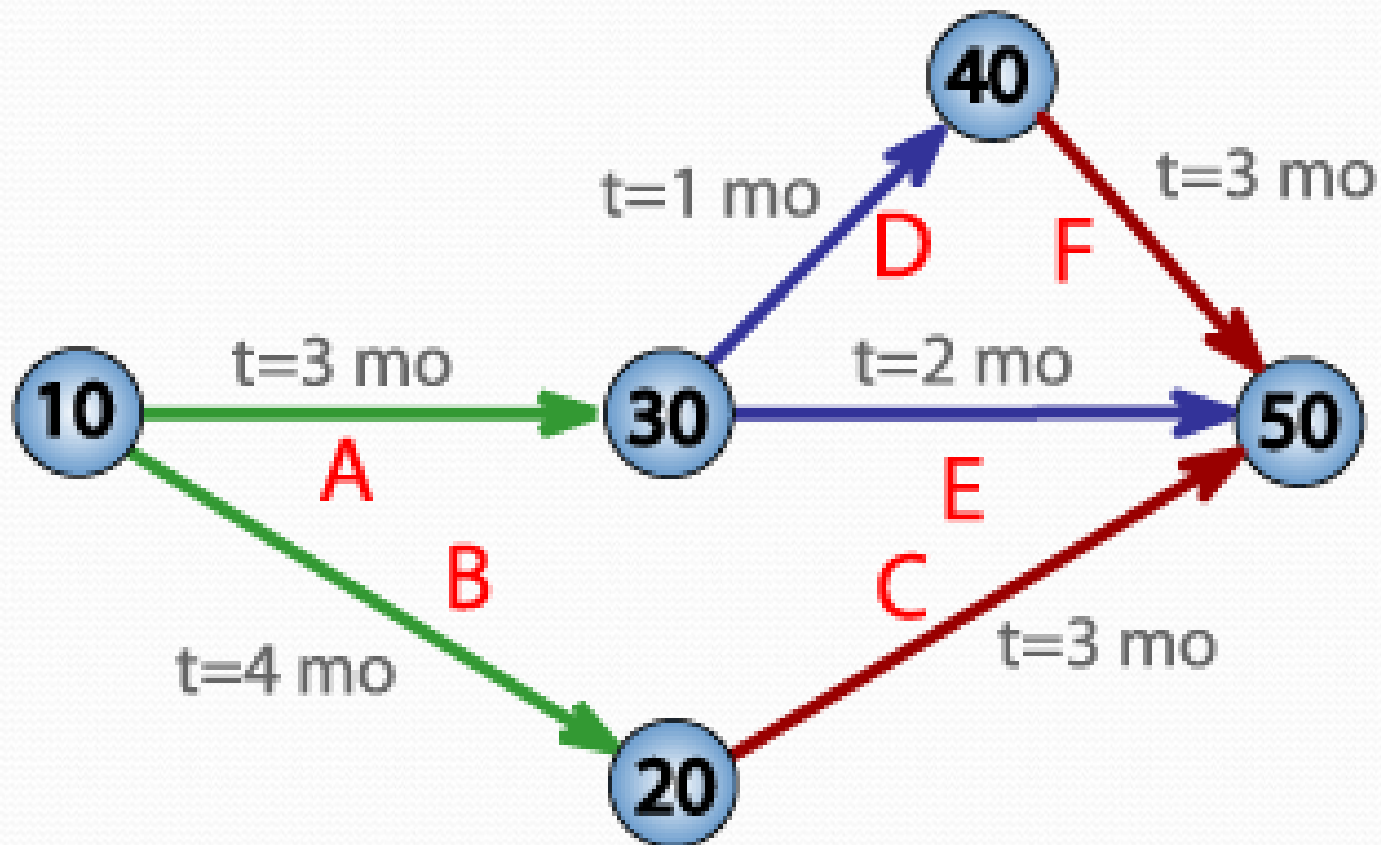
- **Nodes** are visual representations of milestones or events within the project. They are drawn as either numbered boxes or numbered circles.
- **Arrows** are visual representations of the tasks that occur throughout the project. The direction of the arrow indicates the sequence of the task. Diverging arrows show that various tasks can be completed at the same time.

- **Fast tracking** is when tasks and activities are performed simultaneously.
- A **PERT event** is the point at which one or more tasks are started or completed.
- A **predecessor event** occurs immediately *before* some events. A **successor event** naturally occurs *after* events.
- **Slack** is the amount of time a single task can be delayed without harming other tasks or the project as a whole.

- The **critical path** is the longest -- or most time-consuming -- path from the start to the completion of an event or task.
- **Critical path activity** refers to a task that does not experience slack.
- **Crashing critical path** is when the completion time of a task is shortened.
- **Lag time** refers to the earliest point at which a task can follow another.
- **Lead time** is the amount of time it should take to complete a task without impacting the following activities.

- **Expected time** is the best estimation of how long a task will take to complete, taking into consideration any problems or obstacles that might arise.
- **Optimistic time** refers to the minimum amount of time it will take to complete a task.
- **Pessimistic time** is the maximum amount of time it will take to finish a task.
- **Most likely time** is the best guess of how long a task will take, assuming no problems arise.

PERT Chart



Role of a Project Manager

- 
- Project issues
 - Disseminating project information
 - Mitigating project risk
 - Quality
 - Managing scope
 - Metrics
 - Managing the overall work plan

Process
Responsibilities

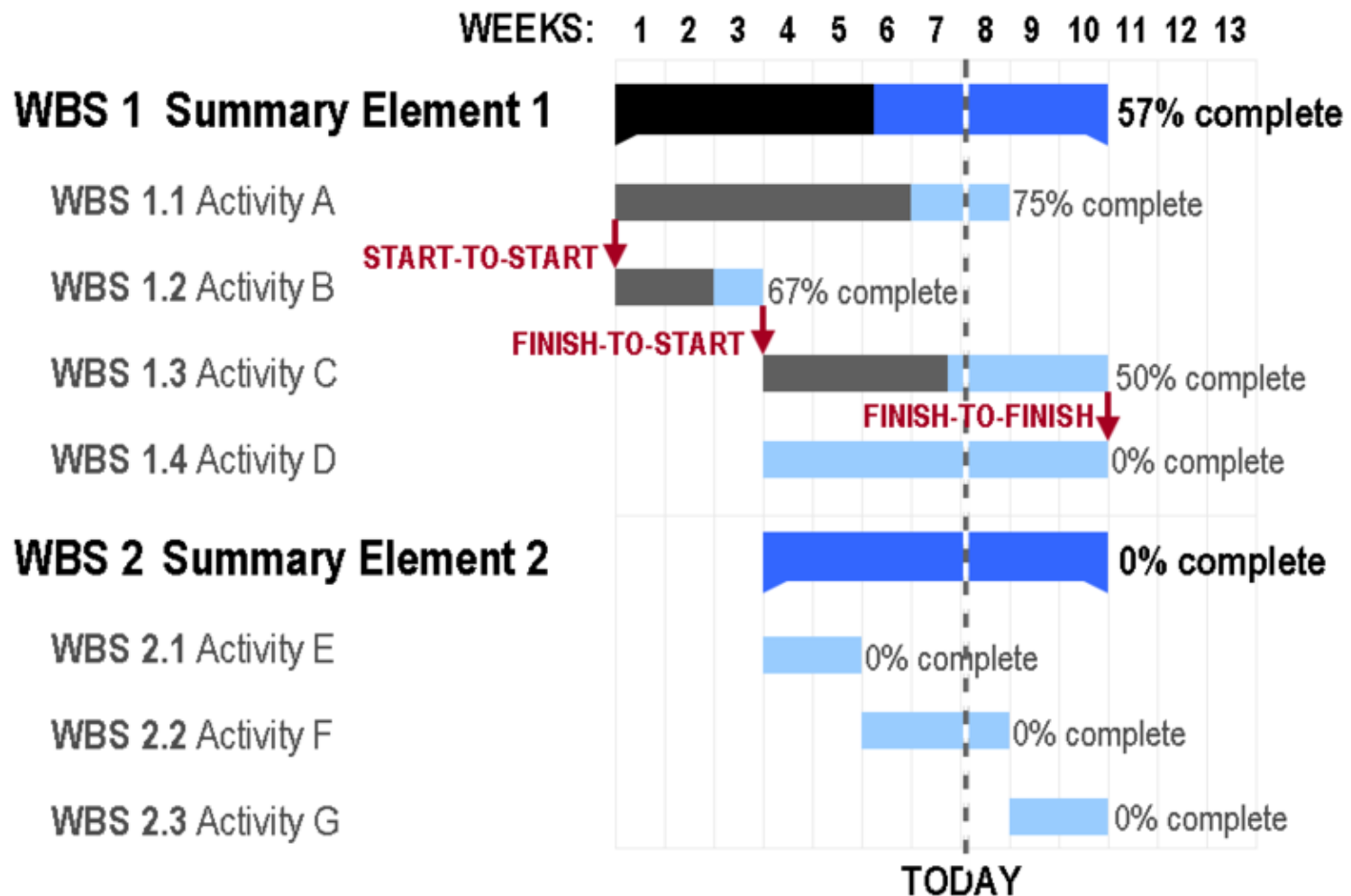
- Implementing standard processes
- Establishing leadership skills
- Setting expectations
- Team building
- Communicator skills

People
Responsibilities

- A Gantt chart is a bar chart that provides a visual view of project tasks scheduled over time. A Gantt chart is used for project planning: it's a useful way of showing what work is scheduled to be done on specific days. It helps project managers and team members view the start dates, end dates and milestones of a project schedule in one simple stacked bar chart.

- A Gantt chart is a stacked bar chart that contains project tasks on a vertical axis and timelines that represent task duration on a horizontal axis. Summary elements, task dependency relationships and milestones in the project schedule are all depicted. The Gantt chart is named after Henry Gantt who popularized this project management chart in the early 20th century.

Gantt Chart



Scope Management

- Project Scope Management is the process to ensure that the project is inclusive of all the work required, and only the work required, for successful completion.
- Primarily it is the definition and control of what *IS* and *IS NOT* included in the project.

Issue Management

- Issues are restraints to accomplishing the deliverables of the project.
- Issues are typically identified throughout the project and logged and tracked through resolution.
- In this section of the plan the following processes are depicted:
 - Where issues will be maintained and tracked
 - The process for updating issues regularly
 - The escalation process
 - The vehicle by which team members can access documented issues

Cost Management

- This process is required to ensure the project is completed within the approved budget and includes:
 - ***Resource Planning*** - The physical resources required (people, equipment, materials) and what quantities are necessary for the project
 - ***Budget***
 - Budget estimates
 - Baseline estimates
 - Project Actuals

Quality Management

- Quality Management is the process that insure the project will meet the needs via:
 - Quality Planning, Quality Assurance, and Quality Control
 - Clearly Defined Quality Performance Standards
 - How those Quality and Performance Standards are measured and satisfied
 - How Testing and Quality Assurance Processes will ensure standards are satisfied
 - Continuous ongoing quality control

Communications Management

- This process is necessary to ensure timely and appropriate generation, collection, dissemination, and storage of project information using:
 - Communications planning
 - Information Distribution
 - Performance Reporting
- Define the schedule for the Project Meetings (Team, OSC, ESC), Status Meetings and Issues Meetings to be implemented

Risk Management

- Risk identification and mitigation strategy
- When\if new risks arise
- Risk update and tracking

Change Control Management

- Define how changes to the project scope will be executed
 - Formal change control is required for all of the following
 1. Scope Change
 2. Schedule changes
 3. Technical Specification Changes
 4. Training Changes
- All changes require collaboration and buy in via the project sponsor's signature prior to implementation of the changes